NISHANT PARWANI

BTech Computer Engineering + MBA Candidate | NMIMS Mumbai
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RESEARCH AND ACADEMIC PROFILE

I am a dual-degree student at NMIMS Mumbai pursuing BTech Computer Engineering and MBA simultaneously (2024-2029). My academic and project work spans machine learning model development, IoT and embedded systems engineering, blockchain application design, and data-driven economics research. I am also a practicing entrepreneur, having co-founded Tesseract Solutions, a B2B technology services startup. I am an active contributor to college governance as Class Representative and Student Council Contingent Leader, and to the research community as an undergraduate author of an independently conducted economics preprint published on SSRN.

My research interests span the intersection of machine learning, applied statistics, and real-world systems particularly in health technology and financial prediction. I approach both technical and strategic problems through an evidence-driven methodology rooted in formal academic training.

EDUCATION

Dual Degree: BTech Computer Engineering + MBA — NMIMS (Narsee Monjee Institute of Management Studies), Mumbai *2024 – 2029*

- CGPA: 8.43 / 10 after First Year
- Class Representative and Student Council Contingent Leader — managing inter-college delegation logistics and peer coordination for 100+ students
- Pursuing undergraduate research in technology and economics; one SSRN-published preprint to date
- Core coursework areas: Data Structures and Algorithms, Python Programming, Business Economics, Entrepreneurship, Financial Accounting, Organizational Behavior, Marketing Management

Class XII — CBSE — Udgam School for Children, Ahmedabad *Graduated 2024*

- 78% aggregate (Physics, Chemistry, Mathematics)
- Class X: 87% | SAT Score: 1340

RESEARCH AND PUBLICATIONS

Published Preprint

Undergraduate Economics Research Paper — SSRN (Social Science Research Network) *2025*

- DOI: dx.doi.org/10.2139/ssrn.6289139
- Conducted as independent undergraduate research without faculty supervision, demonstrating self-directed research capability

Methodology Overview

- Research Question Formulation: Identified a gap in applied economics literature relevant to Indian economic conditions and framed a hypothesis derived from review of existing SSRN and peer-reviewed sources.
- Literature Review Process: Systematically reviewed 15+ prior academic works across economics and policy journals; found prior findings to identify conflicting evidence and theoretical gaps that motivated the study.
- Results Interpretation: Interpreted regression coefficients with appropriate confidence intervals and p-values; discussed practical versus statistical significance; acknowledged limitations including potential omitted variable bias and data granularity constraints.
- Dissemination: Submitted and published on SSRN — the world's leading preprint repository for social science research — achieving open-access indexing across the platform's global research network.

Research Poster Presentation

Inter-College Research Poster Symposium — NMIMS Mumbai *2025*

- Secured 2nd place among 80 participants; communicated complex technical research findings to a multidisciplinary academic panel
- Demonstrated ability to translate quantitative methods into accessible visual and verbal narratives for non-specialist audiences

LEADERSHIP AND EXPERIENCE

Technical Volunteer — Sant Hirdaram Swami Basantram Sewa Trust *May – Jun 2025*

- Deployed digital outreach solutions to extend the Trust's media visibility and beneficiary reach
- Managed registration and data management systems for 100+ beneficiaries; improved operational record accuracy and information accessibility

Public Relations Sub-Head — 4C – The Marketing Cell, NMIMS *Jun 2025 – March 2026*

- Progressed from Volunteer PR Executive (Dec 2024) to Sub-Head within six months, reflecting measurable performance and initiative
- Generated 2,000+ event registrations for flagship college fest across Wings and Roots event verticals through strategic outreach and stakeholder communication campaigns
- Mentors junior PR executives and oversees external brand communication for one of NMIMS's most active student chapters

Outreach Core Member — Taqneeq TechFest (TQ 18.0), NMIMS *Aug 2025 – Feb 2026*

- Promoted from Logistics Executive (TQ 17th Edition) to Outreach Core for TQ 18.0 based on demonstrated execution quality
- Drove external partnerships, sponsorship acquisition, and inter-college participant recruitment for one of Mumbai's largest annual tech festivals

Student Council Contingent Leader and Class Representative — NMIMS MPSTME *2024 – Present*

- Elected to represent MPSTME at inter-college delegation events; coordinated logistics for 100+ student contingents across multiple fests
- Manages student-faculty liaison responsibilities and serves as first escalation point for academic and administrative issues

Outreach Executive — Sattva Cultural Fest, NMIMS *Oct 2024 – May 2025*

- Organized and hosted the Contingent Leaders' Meet for MPSTME, coordinating cross-college participation and inter-delegation communication

TECHNICAL PROJECTS IN DETAIL

Machine Learning and Data Science

Land Price Predictor — Python, scikit-learn, XGBoost, pandas

- Implemented ensemble regression pipeline: Linear Regression as baseline, Random Forest for non-linear feature interactions, and XGBoost for gradient-boosted tree performance
- Applied real-world property datasets to demonstrate practical machine learning engineering methodology

Stock and Finance Predictor — Python, APIs, Time-Series ML

- Integrated live financial data APIs to construct automated data ingestion pipeline with preprocessing and normalization layers

Mumbai Suburban Transport Survey Analysis — R,python, pandas, NumPy

- Designed and executed data-driven survey study of Mumbai's suburban rail network; cleaned and aggregated multi-variable survey data
- Derived actionable policy insights and visualized findings using descriptive statistics and comparative analysis

IoT and Embedded Systems

Real-Time Cardiac Abnormality Detection System — ESP32, C++, IoT

- Engineered live heartbeat monitoring system on ESP32 microcontroller with real-time signal acquisition, threshold-based abnormality detection, and alert notification pipeline
- Integrated sensor data acquisition, on-device signal processing, and IoT connectivity in a single embedded deployment demonstrating applied health-tech engineering

Gesture-Controlled Robot with Android Interface — Arduino, APDS-9960, Android

- Integrated APDS-9960 proximity and gesture recognition sensor (RGB, gesture, ambient light via I2C) with Arduino microcontroller; debugged I2C address conflicts and interrupt handler timing at register level
- Developed companion Android APK interface for wireless gesture command transmission; implemented hardware-software communication bridge for autonomous robot navigation
- Troubleshooting involved logic analyzer verification of I2C signal integrity, power rail noise isolation, and firmware interrupt debouncing — demonstrating systematic embedded systems debugging methodology

SONAR-Based Radar System — Arduino, Processing, Ultrasonic Sensors

- Applied HC-SR04 ultrasonic sensor with servo-motor sweep to implement real-time proximity radar with Processing-based graphical visualization
- Implemented signal processing pipeline: distance calculation from time-of-flight measurements, angular mapping, and real-time display rendering

Blockchain-Inspired Secure Voting Machine — SHA-256, Firebase, Python

- Implemented SHA-256 hashing to create tamper-proof vote records; each vote entry generates a hash chained to the prior entry, applying distributed ledger integrity principles
- Integrated Firebase real-time database for persistent, cloud-synchronized storage of voting records with read/write access controls

Full-Stack and Software Engineering

Python Student Portal — CRUD Database System — Python, SQLite3

- Designed relational schema and implemented full CRUD operations (Create, Read, Update, Delete) using SQLite3 with Python interface
- Built CLI-based multi-user interface supporting concurrent data lifecycle management for student records

Personal Portfolio Website — HTML, CSS, JavaScript, Three.js, GSAP

- Designed and deployed nishantparwani.com with responsive layout, Three.js 3D animations, and GSAP scroll-transition effects
- Implemented mobile-responsive grid layout and cross-browser compatibility testing

CERTIFICATIONS AND FORMAL TRAINING

Entrepreneurship and Design Thinking — IEEE / Formal Coursework

- Completed structured entrepreneurship training grounded in design thinking methodology, covering the full five-phase iterative cycle:
  - Empathize: User interviews and ethnographic observation to identify genuine pain points and unmet needs; avoided assumption-driven problem framing
  - Define: Synthesized user research data into a precise problem statement (Point-of-View statement and How-Might-We questions)
  - Ideate: Applied structured brainstorming techniques (SCAMPER, mind mapping, worst possible idea inversion) to generate a broad solution space before converging on high-potential concepts
  - Prototype: Rapid paper and digital prototyping to make ideas tangible for user feedback; scope limited to testable hypotheses rather than full builds
  - Test: Structured user testing with observation protocols, qualitative feedback synthesis, and iteration cycles — directly applied to Tesseract Solutions' product development process
- Additional modules covered: Business Model Canvas construction, Total Addressable Market sizing, competitive analysis frameworks (Porter's Five Forces, Blue Ocean Strategy), investor pitch structuring, and unit economics modeling

TECHNICAL SKILLS AND COMPETENCIES

Programming Languages
Python (primary), C++, Java, JavaScript, HTML, CSS, MySQL, R (working knowledge)

Machine Learning and Data Science
TensorFlow, XGBoost, scikit-learn (Random Forest, Logistic Regression, Linear Regression), pandas, NumPy, RegEx, data preprocessing pipelines, feature engineering, model evaluation (RMSE, R-squared, cross-validation), time-series forecasting

Embedded Systems and IoT
ESP32, Arduino (C++ firmware), APDS-9960 gesture and proximity sensor (I2C protocol), HC-SR04 ultrasonic sensor, servo motor control, Firebase real-time database, SHA-256 cryptographic hashing, Android APK integration, I2C debugging, interrupt handling

Full-Stack and Web Development
HTML5, CSS3, responsive design, JavaScript (DOM manipulation), Three.js (3D graphics), GSAP (animation) (working knowledge), SQLite3, AngularJS (basics), Firebase (NoSQL cloud database), portfolio deployment

Business and Research Tools
Bloomberg Terminal, MS Office Suite (PowerPoint, Word), AutoCAD, OLS regression, academic publishing, business model design, market validation frameworks

Languages Spoken
English (fluent), Hindi (native), Gujarati (native), Sindhi (native), French (A2)

AWARDS AND RECOGNITION

Smart India Hackathon — 2nd Place, Health and Technology (Round 2) — National Competition

- Developed working healthcare prototype under time-constrained hackathon conditions in a nationally competitive Machine Learning and Software Engineering environment
- Demonstrated ability to ideate, prototype, and present an end-to-end technical solution addressing real healthcare infrastructure challenges within a compressed timeline

Inter-College Research Poster Symposium — 2nd Place — NMIMS *2025*

- Ranked 2nd among 80 participants; evaluated by multidisciplinary faculty panel on research quality, methodology rigor, and communication clarity

Model United Nations — Verbal Commendation and Special Mention (x2) — Multiple Editions *2022–24*

- Recognized for analytical reasoning, diplomatic argumentation, and public speaking in formal parliamentary debate settings
- Demonstrates transferable skills in structured argumentation, evidence synthesis, and persuasive communication directly relevant to research presentations and stakeholder engagement

EXTRACURRICULAR ACTIVITIES

- Competitive Tennis Player: Active participation at competitive level; develops discipline, performance under pressure, and physical fitness
- Polyglot Communication: Proficiency across six languages enables effective cross-cultural communication and collaboration in diverse academic and professional settings

CONTACT AND LINKS

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